



SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE CODE : CSA40

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COURSE NAME : MANAGEMENT INFORMATION SYSTEM

Reg NO: 192110204

9) A college has more than thousand security persons, who are instructed to give duties at different places within the campus. Additionally, they also maintain a routine, which contains all information, such as Date, Duty Start Time, Duty End Time, and Place. Most importantly, all the places are covered by at least one security person. If a security person takes leave, manual entry is done against that person. Finally, at the end of a month, the security persons get paid for their duties, while considering the number of leaves as well. You can see that the manual calculation/operation is a heavy task for the security manager. Therefore, the objective is to build an Online security management system using class diagram through which entire security system within the campus can be controlled in an efficient manner .

Aim:

To design an **Online Security Management System** using a **Class Diagram**, which automates the **duty assignment, leave tracking, and payroll calculation** for security personnel within the college campus.

Procedure:

1. System Login & Authentication

- The security manager and staff log in to access the system.

2. Security Personnel Registration

- Each security person is added to the system with personal details, shift schedules, and assigned duty locations.

3. Duty Allocation & Scheduling

- The system automatically assigns security personnel to different places based on availability and requirements.

4. Routine & Shift Management

- The system maintains a routine with Date, Duty Start Time, Duty End Time, and Duty Location.

5. Leave Management

- Security personnel can request leave, and manual entry for absentees is recorded.

6. Automatic Duty Adjustment

- When a security person is absent, the system suggests replacements to ensure coverage.

7. Attendance Tracking

- The system logs daily attendance and duty completion.

8. Salary Calculation

- Based on working hours and leaves taken, the system calculates salaries automatically.

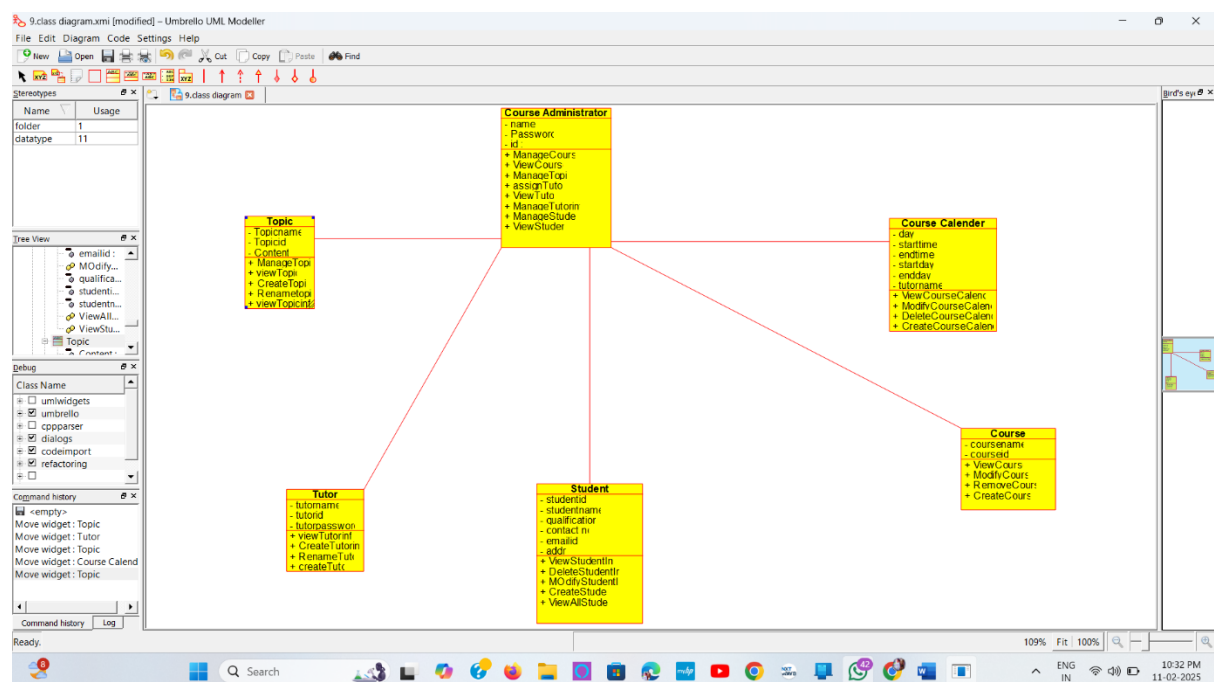
9. Report Generation

- Monthly reports on duty performance, leave records, and salary details are generated.

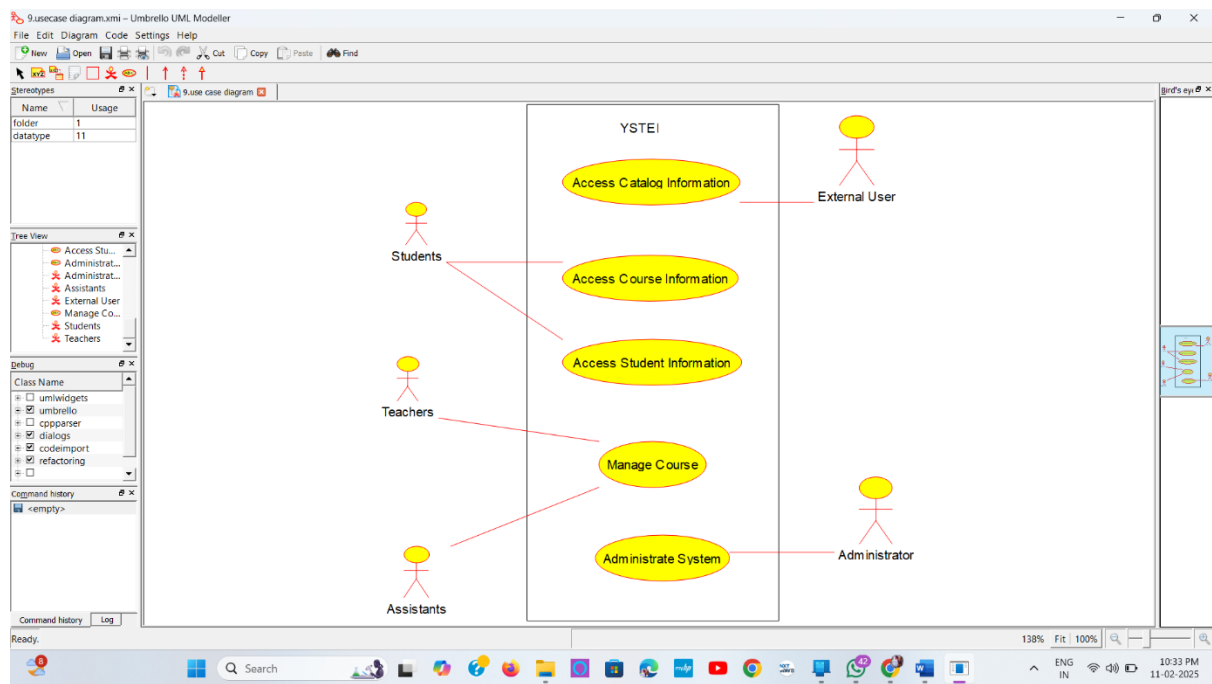
10. System Logout & Maintenance

- Users log out, and administrators ensure the system runs smoothly.

Class Diagram...



Use Case Diagram...



Result:

The **Online Security Management System** improves **efficiency by automating duty scheduling, leave tracking, and salary calculations**. This reduces **manual work, eliminates errors, and enhances overall security management** on the campus.